

Purpose of the Study

The purpose of this research is to evaluate the effectiveness of two different approaches to practicing programming on mobile devices during bus commuting: Parsons Problems and traditional programming problems. As mobile devices become a common tool for students, it is important to understand how different types of exercise impact learning outcomes, usability, workload, and confidence in mobile contexts. By focusing on commuting environments, the research will also examine whether ergonomics affect students differently when using different types of exercise.

Research Questions

1. While commuting, do Parsons problems or traditional programming require less cognitive load?
2. While commuting, do Parsons problems offer better ergonomics compared to traditional programming?
3. While commuting, do Parsons problems and/or traditional programming increase student confidence?
4. While commuting, do students prefer to use Parsons problems or traditional programming?

Experiment Steps:

1. All the students will solve the practice problem during their commute by bus. We will divide them into two groups.
 - a. The first group will solve traditional programming first, then the parson's problem.
 - b. The second group will solve the parson's problem first, then the traditional programming questions.
2. After each practice session, students will fill out survey questions regarding CMDQ, UMUX_Lite, CS Cognitive Load Component Survey.
3. After both the sessions the student will answer the open-ended survey questions along with their demographic information at their home.